Reliable Machine Learning with Unreliable Black Boxes **Spring 2026**

Lecture 5: Uncertainty Quantification Day 4

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1 Efficiency of Prediction Sets

Recap. In the previous lectures, we designed techniques that used a few calibration samples to give a prediction set which is statistically valid i.e. satisfies *coverage*.

Formally, given calibration samples $(X_1, Y_1), \ldots, (X_n, Y_n), X_{n+1}$, let $C(X_{n+1}) = \{y \in \mathcal{Y} \mid s(X_{n+1}, y) \leq \hat{q}_{1-\alpha}\}$ where s is a score function and $\hat{q}_{1-\alpha}$ is the appropriate quantile value on the scores of the calibration samples.

This guarantees (marginal) coverage:

$$\Pr[Y_{n+1} \in C(X_{n+1})] \geq 1 - \alpha$$

The above formulation does not use anything about how the score function looks like, or how they were computed; it was treated as a black box.

Efficiency of Sets. Different choices of the score function s leads to different prediction sets, with varying sizes. Ideally, we would want our prediction set $C(X_{n+1})$ to be as small as possible (e.g. in volume), while satisfying coverage. We call this *efficiency* (or volume optimality/size optimality, etc.).

To understand this better, we start with a simpler, unsupervised setting where the labels are independent of the features.

2 Unsupervised Setting

Here, we are just given $Y_1, Y_2, \ldots, Y_n \sim P$ i.i.d, where P is a probability distribution over the reals. Our goal is to output a set $\widehat{C}$ which simultaneously satisfies coverage and efficiency:

Input: $Y_1, \ldots, Y_n \sim_{\text{iid}} P$
Goal: Output $\widehat{C} = C(Y_1, \ldots, Y_n)$ satisfying

$$\Pr_{Y_{n+1} \sim P}[Y_{n+1} \in \widehat{C}] \geq 1 - \alpha, \qquad \text{(coverage)}$$

$$(\widehat{C}) \leq \beta \cdot \min_{C \subseteq \mathbb{R}}(C). \qquad \text{(efficiency)}$$

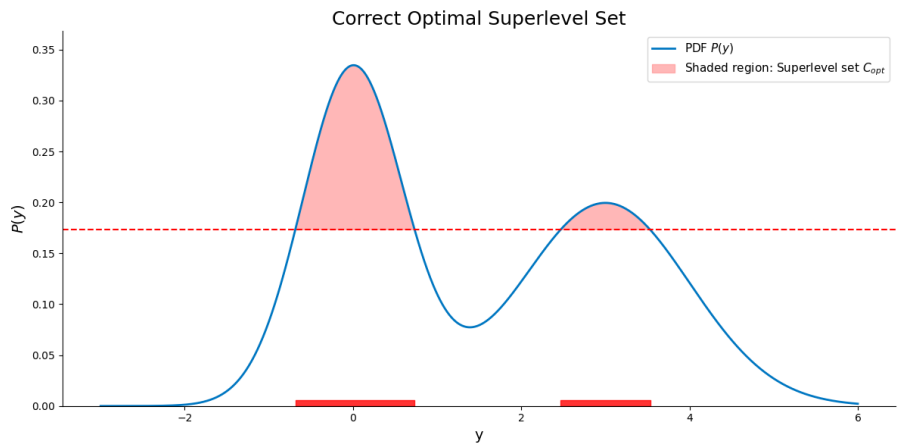

Figure 1: The optimal set that satisfies coverage.

In the above formulation, $\beta \geq 1$ is the approximation ratio - ideally, we would want β to be very close to 1, but sometimes this might be unavoidable. Similarly, we might allow a small slack in what we cover compared to what the optimal coverage set does.

Now, suppose we knew the density function for P. Then, we are essentially looking for the smallest superlevel set of P that cuts off $1 - \alpha$ mass i.e. the line parallel to the x-axis that cuts off $1 - \alpha$ mass.

However, density estimation for general distributions is impossible using finite samples (as described below), especially in higher dimensions. Furthermore, the superlevel sets might be extremely complicated for general distributions, and so might not be particularly useful for applications.

Impossibility of Finding Optimal Volume Set. Consider the standard Gaussian distribution, and sample n points from the distribution. Next, consider the discrete distribution that includes those same n points (and has a slightly larger support, say over $m \geq n^2$ points to avoid collisions).

It is impossible to find the optimal volume set for the above two distributions using just the n samples. However, the two have very different minimum volume sets! Thus, it is statistically intractable to output the optimal volume set, let alone computationally.

Rescuing the Idea. We saw that finding small confidence sets is impossible statistically, in general. At the same time, as stated above, the optimal confidence sets might even be very complicated and not very useful in practice.

A neat idea addresses both the above points: we focus on "simple" prediction sets, and only compete against the best "simple" prediction set!

These simple sets could be intervals, unions of intervals, balls in $\mathbb{R}^d$, product sets, etc. This not only makes the problem tractable, but also makes the outputs practically useful!

3 Finding Small Confidence Sets

We formalize the notion of "simple" by using the concept of VC dimension. The VC dimension can be thought of as a measure of complexity for $\mathcal{F}$. As a rough rule of thumb, it matches the parameters needed to specify a set in the family e.g. the family of intervals has a VC-dimension of 2, unions of k-intervals has a VC-dimension of $2k$, balls in $\mathbb{R}^d$ has a VC-dimension of $d+1$, and so on.

We now seek to solve the following problem.

Input: $Y_1, \ldots, Y_n \sim_{\text{iid}} P$, a family of sets $\mathcal{F}$
Goal: Find

$$\min_{C \in \mathcal{F}} (C)$$
$$\text{s.t. } P[C] \geq 1 - \alpha.$$

The family of sets having a bounded VC-dimension implies the statistical tractability of finding minimum volume confidence sets!

Theorem 1 (Uniform Convergence)**.** *Let $\mathcal{F}$ be a family of sets with VC-dimension k. Given n i.i.d samples $Y_1, \ldots, Y_n \sim P$, where P is an* arbitrary *distribution, we have, with probability 0.99,*

$$\sup_{C \in \mathcal{F}} \left| P(C) - \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}[Y_i \in C] \right| \leq O\left(\sqrt{\frac{k}{n}}\right)$$

Thus, you sample $n = O\left(\frac{k}{\epsilon^2}\right)$ points, look at all $C \in \mathcal{F}$ that satisfies $\frac{1}{n}\sum_{i=1}^{n} \mathbb{I}[Y_i \in C] \geq 1 - \alpha - \epsilon$, and output the minimum volume one. This would guarantee, by Theorem 1, that $P[C] \geq 1 - \alpha$.

Extension to Supervised Setting. We can extend the above to the setting with features, by using the distribution conformal prediction setting from the last lecture to approximate the conditional expectation: $\widehat{F} \approx F(Y|X)$, and find the minimum volume prediction set for $Y|X$. This would admit the following guarantees.

Theorem 2. *The above admits the following guarantees:*

- *It achieves marginal coverage always.*
- *If $\widehat{F}$ is δ-close to $F(Y|X_{n+1})$, we get conditional coverage upto $O(\delta)$.*
- *It achieves volume optimality against sets from $\mathcal{F}$.*

Computational Tractability? The above just guarantees only statistical tractability. In general, it can be computationally intractable even for sets with a bounded VC-dimension. However, we can get efficient results for the set of intervals, unions of intervals, and some other classes. Furthermore, we can also get approximation guarantees for balls/ellipsoids in $\mathbb{R}^d$.

4 Score Functions to Prediction Sets

So far, we have seen how to use score functions to get prediction sets. We can also go the other way - use prediction sets to get score functions.

For this, suppose we have a laminar family/nested family of prediction sets $C_1 \subseteq C_2 \subseteq \cdots \subseteq C_m$, and it is guaranteed that for some $j^* \in [m]$, $C_{j^*} = C^*$, the optimal prediction set. Then, let $s^*(y) := \frac{1}{m}\sum_{j\in[m]} \mathbb{I}[y \notin C_j]$. Then, the above admits the following guarantee.

Theorem 3. *Given $2n+1$ iid observations $Y_1, \ldots, Y_{2n+1}$, using split conformal prediction with s^* gives us a set $\widehat{C}$ s.t.*

- $\Pr[Y_{n+1} \in \widehat{C}] \geq 1 - \alpha$.
- *if training samples are drawn iid from P to produce the volume optimal prediction set C^* used to generate s^*, then, if $n = O\left(\frac{k}{\epsilon^2}\right)$,*

$$(\widehat{C}) \leq \min_{Cs.t.P(C)\geq 1-\alpha+\epsilon} (C)$$